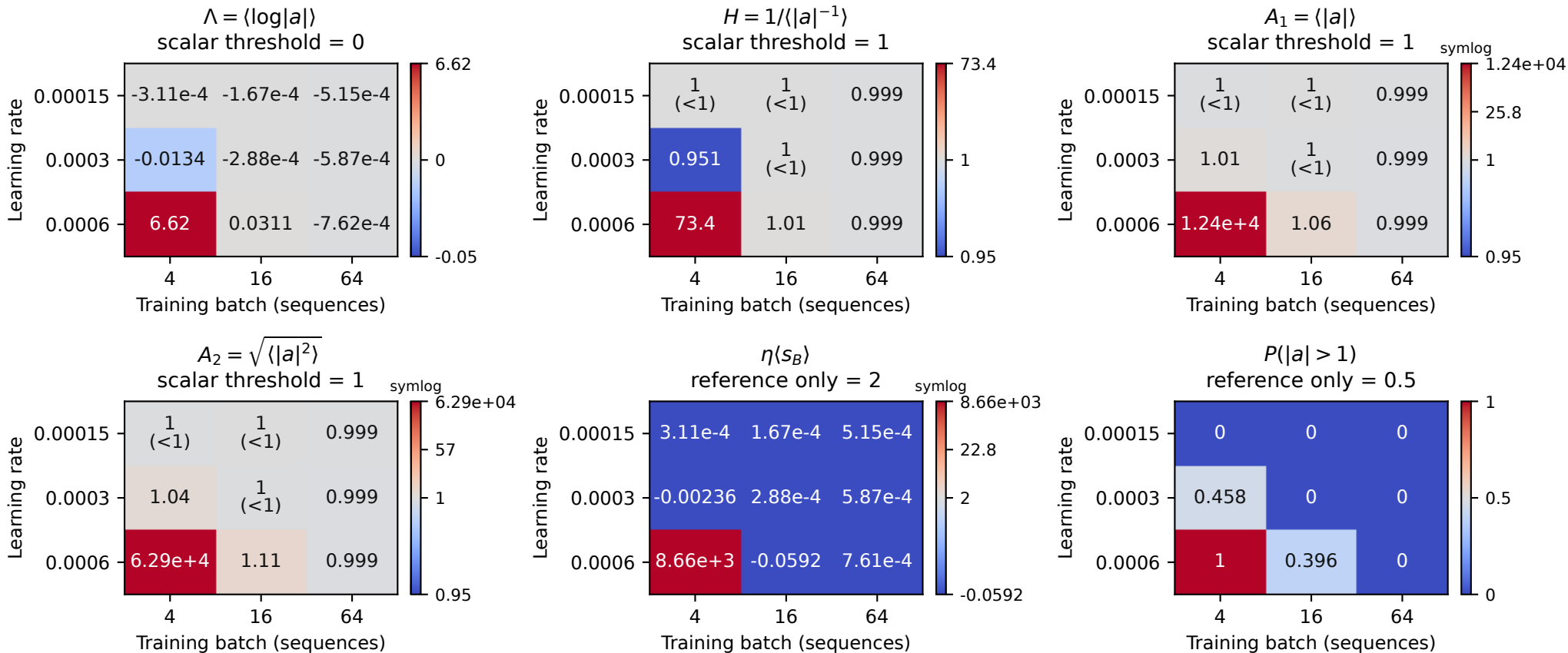


GPT 168M / FineWeb-Edu — Raw Hessian

Training batch $\times$ learning rate; final three checkpoint pools; average run-level diagnostics



Per-run pools: 48 gains at 8.192, 12.288, 16.384M tokens.

Blue / red: below / above reference. Rounded boundary values show their strict side.